

Appendix B

Non-Hamiltonian dynamics

A systematic procedure for extending the techniques of classical statistical mechanics to non-Hamiltonian systems was proposed by Tuckerman et al. [267,715,716]. In the present appendix, we sketch the general approach for analyzing extended Lagrangian systems. We will, however, skip most of the derivations. For a more complete and more rigorous derivation, using the mathematical techniques of differential geometry, the reader is referred to the original references.

In general, the dynamics that results from solving non-Hamiltonian equations of motion is not area-preserving. As we have seen in Appendix A.4.3, solving the equations of motion can be considered as a coordinate transformation from the phase-space coordinates at time t_0 to those at time t . If the system is Hamiltonian, the time evolution of the system will change the shape of an infinitesimal volume element in phase space, but not its volume $d\Gamma$. In contrast, for a non-Hamiltonian system, we have to take into account the Jacobian of the transformation associated with the evolution of $d\Gamma(t_0) \rightarrow d\Gamma(t)$:

$$d\Gamma_t = J(\Gamma_t; \Gamma_0) d\Gamma_0,$$

where the subscript 0 indicates the phase-space volume at $t = t_0$ and J is the determinant of the Jacobian matrix $\mathcal{M}$ of the transformation. For convenience, we choose $t_0 = 0$.

The motion in phase space of a Hamiltonian system is similar to that of an incompressible liquid: in time the volume of this “liquid” does not change. In contrast, a non-Hamiltonian system is compressible. This compressibility must be taken into account when considering the generalization of the Liouville theorem to non-Hamiltonian systems.

The compressibility can be derived from the time dependence of the Jacobian

$$\frac{dJ(\Gamma_t; \Gamma_0)}{dt} = \kappa(\Gamma_t, t) J(\Gamma_t; \Gamma_0) \quad (\text{B.0.1})$$

in which $\kappa(\Gamma_t, t)$, the *phase space compressibility* of the dynamical system, is defined:

$$\kappa(\Gamma_t, t) \equiv \nabla_{\Gamma} \cdot \dot{\Gamma}. \quad (\text{B.0.2})$$

Eq. (B.0.1) has as formal solution

$$J(\Gamma_t; \Gamma_0) = \exp \left(\int_0^t \kappa(\Gamma_s, s) ds \right).$$

If we define $w(\Gamma_t, t)$ as the primitive function of $\kappa(\Gamma_t, t)$, then we can write

$$\begin{aligned} J(\Gamma_t; \Gamma_0) &= \exp[w(\Gamma_t, t) - w(\Gamma_0, 0)] \\ &\equiv \frac{\sqrt{g(\Gamma_0, 0)}}{\sqrt{g(\Gamma_t, t)}}, \end{aligned}$$

where the last line defines the quantity $\sqrt{g}$. Recall that

$$\begin{aligned} d\Gamma_t &= J(\Gamma_t; \Gamma_0) d\Gamma_0 \\ &= \frac{\sqrt{g(\Gamma_0, 0)}}{\sqrt{g(\Gamma_t, t)}} d\Gamma_0. \end{aligned}$$

Hence,

$$\sqrt{g(\Gamma_t, t)} d\Gamma_t = \sqrt{g(\Gamma_0, 0)} d\Gamma_0,$$

which defines an invariant measure in phase space. This result can be used to derive a new form of the Liouville equation for non-Hamiltonian systems. The important point here is that the phase-space distribution, $f(\Gamma)$, the function in which we are interested, which gives the probability density in phase space, should be kept separate from the phase-space metric, $\sqrt{g}$, which ensures that the volume of phase space of a non-Hamiltonian system is invariant under time evolution,

$$\frac{\partial f \sqrt{g}}{\partial t} + \nabla \cdot (f \sqrt{g} \dot{\Gamma}) = 0. \quad (\text{B.0.3})$$

The expression corresponding to an ensemble average is

$$\langle A \rangle = \frac{\int d\Gamma \sqrt{g(\Gamma)} A(\Gamma) f(\Gamma)}{\int d\Gamma \sqrt{g(\Gamma)} f(\Gamma)}. \quad (\text{B.0.4})$$

Assuming that there are n_c conservation laws, $\Lambda_k(\Gamma') = C_k$ for $k = 1, \dots, n_c$, the partition function of the non-Hamiltonian system is given by

$$\Omega(C_1, \dots, C_n) = \int d\Gamma' \sqrt{g(\Gamma')} \prod_{k=1}^{n_c} \delta(\Lambda_k(\Gamma') - C_k). \quad (\text{B.0.5})$$

In many applications, one obtains the correct (NVT or NPT) partition function from the above “microcanonical” partition function, by carrying out the integration over the unphysical variables that have been introduced to represent the effect of a thermostat or barostat. In order to do this properly, it is essential

to identify *all* conservation laws. Moreover, it is useful to eliminate from the analysis all those coordinates that are linearly dependent on other variables and variables that are “driven.” Variables are called “driven” when they do not influence the time evolution of (and are not coupled through a conservation law) the physical variables of interest in the system, even though their own time evolution may depend on these last variables.